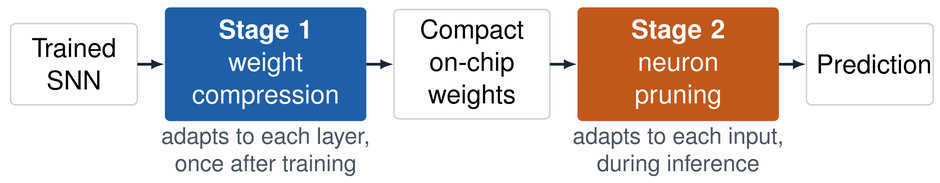
Adaptive Information Compression for Energy-Efficient Spiking Neural Networks

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Introduction

Spiking neural networks (SNNs) suit low-power edge devices, yet much of what they store and compute carries little information:

- **Stored weights.** Models outgrow on-chip memory, and off-chip access adds latency and energy [1].
- **Spiking activity.** Neurons keep updating after their contribution to the output has become small.



Each stage has been evaluated on its own so far. Evaluating the full pipeline is the next step.

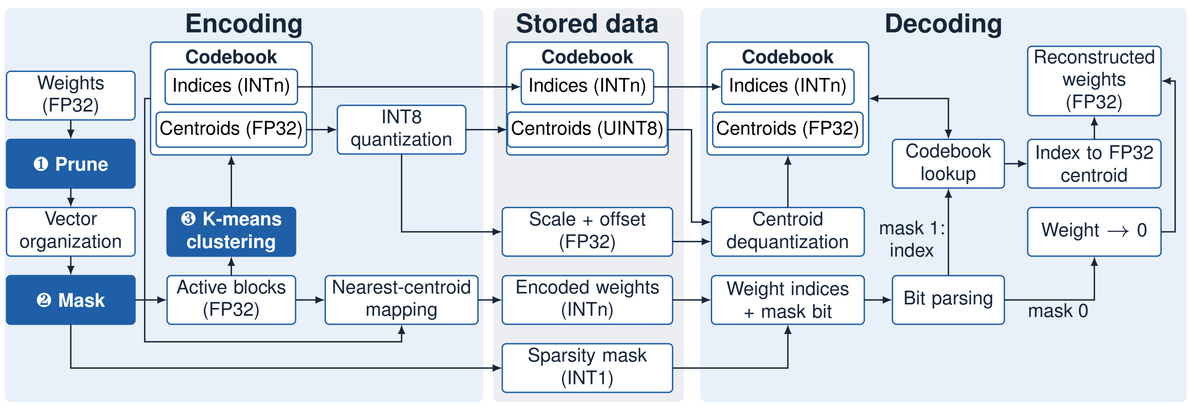
Weight Compression

Problem. Standard vector quantization (VQ) spends centroids on near-zero weights and turns exact zeros into small non-zero values, which accumulate over time as *synaptic leakage* [2].

Approach. Applied layer by layer, with no retraining:

1. **Prune.** Set weights with $|w| \leq \tau$ to exact zero, where $\tau = k\,\Delta W$ and $\Delta W = \max|W|/(2^{b-1}-1)$.
2. **Mask.** One bit per block marks it all-zero or active.
3. **Cluster.** K-means runs on active blocks only, giving a B_{idx}-bit index into an INT8 codebook.

A sparse block costs 1 bit and an active block $1 + B_{\text{idx}}$ bits. On decoding, mask 0 emits zero, so pruned weights stay exactly zero.

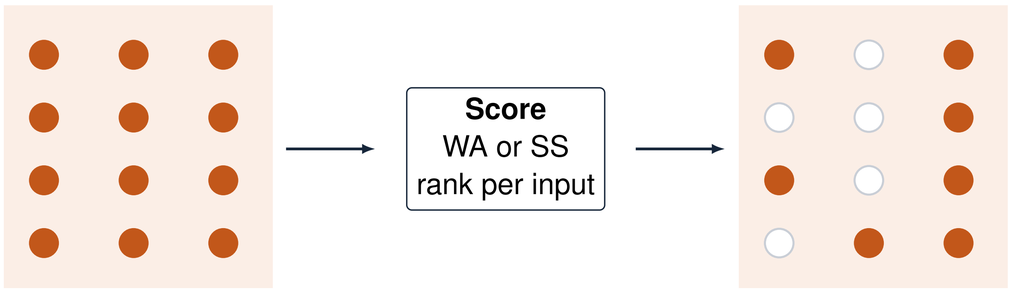


Adaptive Neuron Pruning

Problem. Static pruning fixes the network before deployment, yet a neuron's importance changes with the input as spikes accumulate.

Approach. Observe early spikes for each input, then disable low-contribution neurons for the remaining timesteps:

Phase 1: observe all neurons, $t \leq T_p$ — **Phase 2: continue** survivors only, $t > T_p$



Neurons are ranked from Phase-1 spike counts c_l and outgoing weights:

$$s_l^{\text{WA}} = c_l \odot \|W_{l+1}\|_{2,\text{col}} \qquad s_l^{\text{SS}} = c_l \odot W_{l+1}[\text{winner}, :]$$

WA favours neurons with strong outputs. SS favours neurons that feed the strongest downstream response for this input.

Results

MLP on MNIST

46.55× weight compression
96.82 → 93.00%
2×2, INT5

68.7% est. neuron energy saved
90% pruned after 4/20
96.26 → 95.38%

ResNet-18 on CIFAR-10 [3]

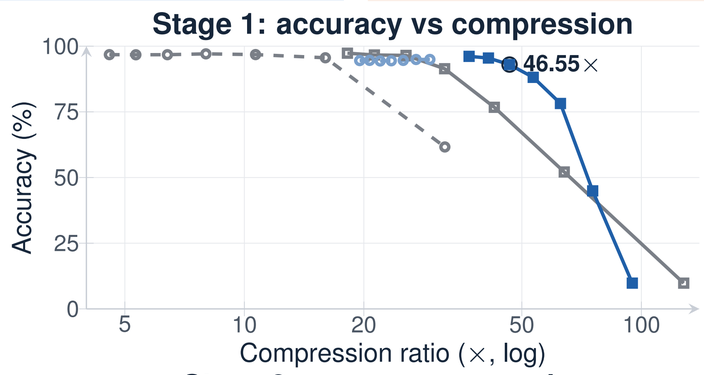
15.37× weight compression
94.09 → 90.51%
1×1, INT3

≤0.2 pt accuracy loss
90% mid-channels pruned, 5/6
92.90 → 92.70% (SS)

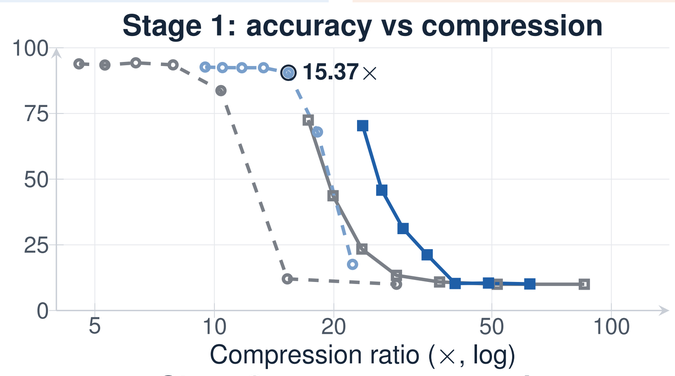
STDP network on MNIST [4]

44.31× weight compression
90.13 → 86.37%
2×2, INT5

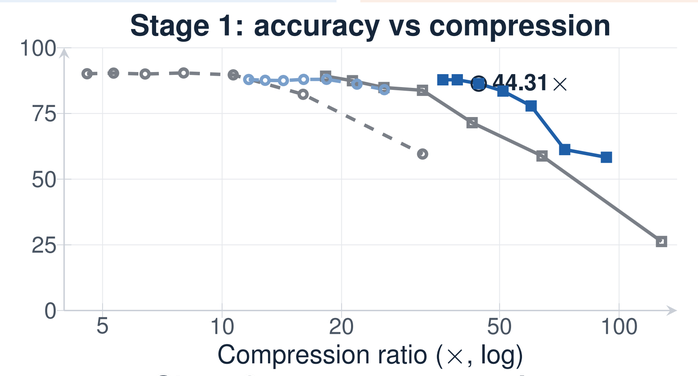
≤0.3 pt accuracy loss
99% pruned after 75/250
92.15 → 91.93% (SS)



Stage 1: 784:1000:10. Stage 2: 784:100:100:10, pruned after 4/20.



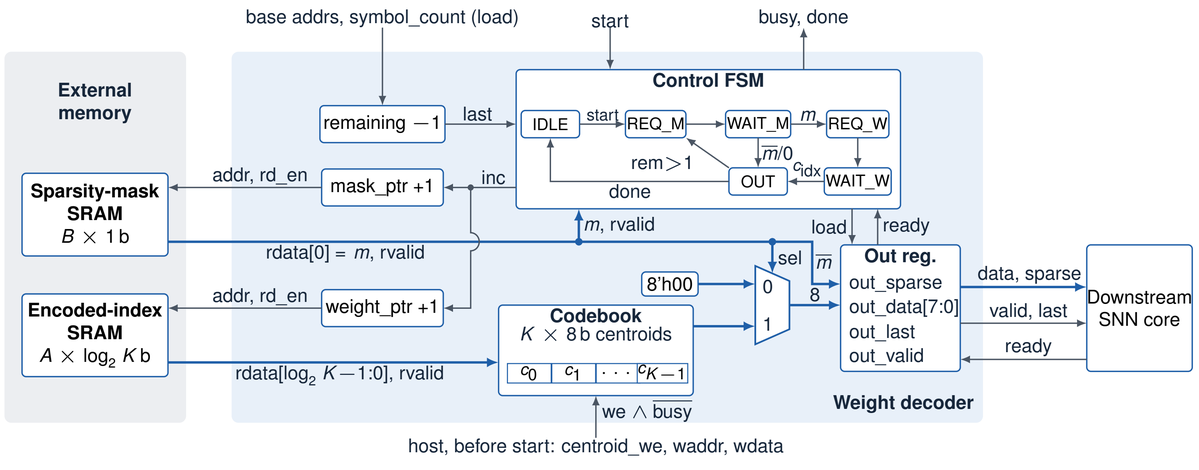
Stage 1: $T = 400$. Stage 2: $T = 6$, SS; labels: timesteps observed. Pruning: intra-block channels only.



Stage 1: 784:1000. Stage 2: 784:500, SS; labels: timesteps observed of 250.

Stage 1 legend: Standard VQ 1×1 — Standard VQ 2×2 — Proposed 1×1 — Proposed 2×2

Weight Decoder



FreePDK45 synthesis: **2.13 GHz**, **1,453.16 μm²**, **2.079 mW**, **953** cells

Energy Estimate

Accuracy (%) and estimated neuron-energy reduction (%) for MNIST MLPs with SS.

Neurons pruned	After 4/20 Acc.	After 4/20 Energy	After 8/20 Acc.	After 8/20 Energy	After 12/20 Acc.	After 12/20 Energy
784:100:10 (unpruned: 95.92%, 1.42 nJ)						
50%	95.04	33.9	95.28	25.5	95.48	17.0
75%	95.00	50.1	95.28	37.6	95.46	25.1
90%	95.06	64.1	95.28	48.1	95.52	32.1
784:100:100:10 (unpruned: 96.26%, 2.81 nJ)						
50%	95.26	36.3	96.00	27.1	96.04	18.0
75%	95.26	54.6	95.98	41.0	96.00	27.3
90%	95.38	**68.7**	96.08	51.5	96.16	34.3

Method. Activity-based model of a clock-gated LIF neuron on FreePDK45: $E_{\text{spike}} = 0.4281$ pJ per spike, $E_{\text{idle}} = 0.4977$ pJ per neuron per timestep.

Conclusion

- **Stage 1:** 15.37–46.55× weight compression with 3–4 points of accuracy loss.
- **Stage 2:** 90% of intra-block channels (CNN) and 99% of hidden neurons (STDP) pruned per input, under 1 point lost.
- **Next:** evaluate both stages together, on FPGA and ASIC, with board-level energy.

References

[1] V. Sze et al., *Proc. IEEE*, vol. 105, no. 12, pp. 2295–2329, 2017.
[2] S. Krithivasan et al., *Proc. ACM/IEEE DAC*, pp. 373–378, 2022.
[3] K. He et al., *Proc. IEEE CVPR*, pp. 770–778, 2016.
[4] P. U. Diehl and M. Cook, *Front. Comput. Neurosci.*, vol. 9, art. 99, 2015.